

Yissum Research Intelligence Report — Healthcare — Week 30, 2026

12 high-potential paper(s) · Healthcare · Weekly · Week 30, 2026 · Generated July 22, 2026

This week's digest highlights groundbreaking advancements poised to transform healthcare and biotechnology. Featured innovations include novel cell and gene therapies for neurological disorders and oncology, alongside next-generation platforms enhancing precision diagnostics and drug discovery. Furthermore, we showcase smart materials for infection control and AI-powered software solutions set to optimize patient stratification and therapeutic design. These technologies offer significant market opportunities by addressing critical unmet needs and improving R&D efficiency.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Tamir Ben-Hur** — Neuroscience, Immunology, Clinical, Synthetic Biology
- **Oren Ram** — Genomics, Diagnostics, Drug Discovery, Clinical
- **Oren Zimhony** — Medical Device, Materials, Clinical
- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Mattan Hurevich** — Drug Discovery, Diagnostics, Materials, Synthetic Biology
- **Dmitry Tselikhovsky** — Drug Discovery, Materials, Synthetic Biology
- **Ayelet N Landau** — Diagnostics, Medical Device, Software/AI, Neuroscience
- **Mor Nitzan** — Drug Discovery, Software/AI, Genomics, Immunology
- **Ofer Mandelboim** — Immunology, Drug Discovery, Clinical
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics
- **Yuval Ramot** — Software/AI, Immunology, Clinical, Drug Discovery
- **Ora Schueler-Furman** — Drug Discovery, Software/AI, Proteomics

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders. 36/50

""Off-the-shelf"" stem cell therapy shows promise for repairing myelin damage."

Main researcher: Tamir Ben-Hur · tamir@hadassah.org.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; The Department of Neurology, The Agnes-Ginges Center for Human Neurogenetics, Hadassah-Hebrew University Medical Center, Jerusalem, Israel. Electronic address: tamir@hadassah.org.il.

Published in Stem cell reports · 2026-06

WHY NOW

Addressing a major unmet need in demyelinating disorders like MS, this scalable therapy offers a clear path to clinical development with broad impact.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

COMMERCIALISATION METRICS

Novelty

6/10

While the use of hESC-derived astrocytes is a sophisticated approach to remyelination, the 'bystander mechanism' and the focus on microglial debris clearance are established concepts in neuroregeneration, making this more of an incremental application of cell therapy than a groundbreaking scientific paradigm shift.

Commercial Potential

8/10

The therapy uses clinical-grade, scalable hESC-derived cells targeting a high-unmet-need market (demyelinating disorders) with demonstrated safety and a clear mechanism of action. The 'clinical-grade' status and scalability suggest it is already positioned for translational development rather than being purely theoretical.

Market Size

9/10

The therapy targets demyelinating disorders (such as Multiple Sclerosis), which represent a massive, high-value patient population with significant unmet medical needs. The mechanism's ability to address multiple facets of remyelination failure suggests a broad, multi-indication platform potential within the neurodegenerative market.

Tech Readiness

5/10

The technology has transitioned from basic lab discovery to 'clinical-grade' manufacturing and has demonstrated efficacy in animal models (lysolecithin-induced), but it remains in the preclinical validation stage rather than undergoing human clinical trials.

IP Strength

8/10

The development of a 'clinical-grade' hESC-derived astrocyte product (AstroRx) suggests a proprietary, scalable manufacturing process and a specific cell composition that is highly patentable. The multi-modal mechanism of action—addressing debris clearance and OPC differentiation—provides multiple layers of therapeutic defensibility and platform potential for various CNS disorders.

[View in Dashboard](#) [Open Paper](#)

2. sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method **35/50** for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

"New single-cell sequencing boosts precision diagnostics and drug discovery."

Main researcher: Oren Ram

Department of Biological Chemistry, Alexander Silberman Institute of Life Sciences, The Hebrew University, 9190401 Jerusalem, Israel.

Published in Nucleic acids research · 2026-04

WHY NOW

Capturing more comprehensive RNA data cost-effectively, this platform tackles a key bottleneck in understanding complex diseases and drug resistance mechanisms.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

COMMERCIALISATION METRICS

Novelty

8/10

The technology represents a significant platform innovation by enabling full-length, non-polyA RNA capture in a scalable droplet format, overcoming a major technical limitation of current industry standards like 10x Genomics. This capability to access the 'dark transcriptome' (ncRNA, histone RNA, etc.) provides a new dimension of biological insight that is highly relevant for precision oncology.

Commercial Potential

8/10

The technology offers a significant, platform-enabling improvement over industry standards (10x Genomics) by capturing non-polyA RNA and providing higher UMI density, which has clear commercial utility for drug discovery and precision oncology. Its scalability and cost-effectiveness position it as a highly licensable tool for large-scale clinical biomarker profiling.

Market Size

8/10

The technology targets the high-value single-cell sequencing market by offering a platform-enabling solution for comprehensive transcriptomics, which is essential for advanced drug discovery and personalized oncology. Its ability to capture non-polyA RNAs and alternative splicing at scale provides a significant competitive edge in the expanding precision medicine and biomarker discovery sectors.

Tech Readiness

4/10

The technology is currently at the proof-of-concept stage, demonstrated through successful validation in a laboratory setting using cancer cell lines. While it shows significant performance improvements over existing methods, it has not yet been transitioned into a standardized, commercially available kit or integrated into clinical diagnostic workflows.

IP Strength

7/10

The technology represents a potentially patentable platform innovation by optimizing droplet-based capture for non-polyA RNA, offering a clear technical advantage over industry standards like 10x Genomics. However, defensibility may be challenged by the crowded landscape of single-cell sequencing chemistries and the difficulty of protecting specific microfluidic or bead-based optimizations once widely adopted.

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3. The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms. 33/50

"Novel coating extends disinfectant power, combating hospital superbugs."

Main researcher: Oren Zimhony · Oren_Z@clalit.org.il.

Infectious Diseases Unit, Faculty of Medicine, Kaplan Medical Center, Hebrew University of Jerusalem, Pasternak St 1 Rehovot, Rehovot, POB1 76100, Israel. Oren_Z@clalit.org.il; Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel.

Oren_Z@clalit.org.il.

Published in Antimicrobial resistance and infection control · 2026-06

WHY NOW

Offers hospitals a scalable, durable solution to reduce multidrug-resistant infections, enhancing patient safety and significantly lowering healthcare costs.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

COMMERCIALISATION METRICS

Novelty

5/10

The research presents an incremental improvement on existing disinfection protocols by using a proprietary coating to extend the efficacy of standard chlorine-based agents. While clinically useful, the innovation is an additive formulation rather than a groundbreaking paradigm shift in antimicrobial science.

Commercial Potential

8/10

The technology demonstrates a clear, additive value proposition to existing hospital disinfection protocols and uses a validated proof-of-concept to address high-value clinical needs (MDRO reduction). The 'proprietary formulation' suggests protectable IP that can be easily commercialized as a specialized chemical adjunct for the healthcare sector.

Market Size

6/10

The technology addresses a high-value niche in infection control (MDRO isolation rooms), but its application as an adjunct to existing protocols suggests a specialized additive market rather than a broad-scale replacement for global disinfection commodities.

Tech Readiness

7/10

The technology has moved beyond lab-scale validation into a controlled quasi-experimental clinical setting with real-world implementation in hospital isolation rooms. It is currently at the stage of clinical validation/pilot testing, positioned just before full-scale commercial rollout or large-scale multi-center trials.

IP Strength

7/10

The use of a proprietary formulation (OxiLast™) to stabilize existing disinfectants suggests a protectable chemical composition or method-of-use patent. However, the efficacy relies on an additive application to standard protocols, which may face moderate defensibility challenges against generic formulations or easier-to-replicate chemical stabilizers.

[View in Dashboard](#) [Open Paper](#)

4. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

"Breakthrough gene delivery system simplifies intracellular genetic targeting."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

WHY NOW

This platform overcomes a major hurdle in nucleic acid therapeutics by enabling cell entry without complex, expensive lipid nanoparticles or viral vectors.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

[View in Dashboard](#) [Open Paper](#)

5. Total Synthesis of Conjugation-Ready Sulfated Red Algae Carrageenan Oligosaccharides for Sensing Applications.

31/50

"Precise seaweed oligosaccharides unlock advanced drug delivery and biosensing."

Main researcher: Mattan Hurevich

Institute of Chemistry and Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Safra Campus, Givat Ram, Jerusalem 9190401, Israel.

Published in Journal of the American Chemical Society · 2026-06

WHY NOW

Manufacturing controlled carrageenan opens doors for highly specific therapeutic targeting and sensitive diagnostic tools, enabling novel material applications.

COMMERCIAL ANGLE

The ability to manufacture highly controlled, homogeneous carrageenan building blocks enables the development of advanced targeted drug delivery systems and highly sensitive biosensors.

COMMERCIALISATION METRICS

Novelty

8/10

The work represents a significant leap in glycan engineering by achieving the first total synthesis of homogeneous carrageenan oligogalactans, overcoming long-standing structural complexity barriers. This provides a platform-enabling methodology for precision sulfated polysaccharides that was previously inaccessible via natural extraction.

Commercial Potential

6/10

The paper provides a high-value chemical platform for creating precise, functionalized building blocks, but the commercial application is currently limited to niche sensing and R&D tools rather than a large-scale industrial material.

Market Size

7/10

The ability to produce homogeneous, precisely sulfated carrageenan oligosaccharides unlocks high-value niche markets in precision drug delivery and advanced diagnostic sensing. However, while the platform is enabling, the total addressable market is currently constrained by the specialized, high-margin nature of synthetic glycans compared to bulk commodity carrageenans.

Tech Readiness

3/10

The work is at a fundamental academic stage, focusing on the complex chemical proof-of-concept for total synthesis rather than scalable manufacturing or functional application testing. While the 'conjugation-ready' nature provides a pathway for future platforms, the technology remains in the basic research/lab-scale phase.

IP Strength

7/10

The development of a specialized, multi-level protecting group hierarchy for total synthesis of complex polysaccharides represents a high barrier to entry and strong potential for composition-of-matter patents. While the synthetic methodology is highly defensible, its commercial impact depends on the successful application of these oligosaccharides in specific high-value sensing or therapeutic markets.

[View in Dashboard](#) [Open Paper](#)

6. Proline-promoted electrophilic dichloromethylation of α -enaminones via stereochemically gated resolution.

30/50

"Sustainable, metal-free synthesis method boosts drug and material production."

Main researcher: Dmitry Tselikhovsky · dmitryt@ekmd.huji.ac.il.

The Institute for Drug Research, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, 9112102, Israel.

dmitryt@ekmd.huji.ac.il.

Published in Nature communications · 2026-04

WHY NOW

This efficient chemical platform offers pharmaceutical and materials industries greener, more cost-effective routes to complex molecular building blocks.

COMMERCIAL ANGLE

This platform provides a more efficient and sustainable way to synthesize complex building blocks used in pharmaceutical manufacturing and advanced materials science.

COMMERCIALISATION METRICS

Novelty

8/10

The research introduces a novel closed-shell, metal-free mechanism for polyhalomethylation that shifts away from traditional radical/carbene pathways. The discovery of topology-driven divergence—where ring size dictates either functionalization or spontaneous cascade aromatization—represents a significant, programmable platform for synthetic methodology.

Commercial Potential

6/10

The paper describes a highly versatile, metal-free synthetic platform that could streamline the production of complex building blocks for medicinal chemistry. However, its commercial value is currently limited to a specialized tool for process chemists rather than a stand-alone product or a transformative platform with broad industrial scalability.

Market Size

6/10

The platform offers valuable versatility for medicinal chemistry and fine chemical synthesis by enabling novel functionalization of building blocks, but its market size is constrained to the niche specialty chemical and R&D reagent sectors rather than a large-scale therapeutic or industrial commodity market.

Tech Readiness

3/10

This is fundamental discovery research focused on mechanistic understanding and new synthetic methodology at the academic lab stage. While it demonstrates a novel chemical platform, it lacks the process optimization, scalability testing, or applied proof-of-concept required for higher TRL scores.

IP Strength

7/10

The discovery of a unique, metal-free, closed-shell mechanistic class offers strong potential for broad composition-of-matter and process patents in medicinal chemistry. However, the utility is currently limited to fundamental synthetic methodology, which may face challenges in demonstrating high-value industrial exclusivity compared to targeted therapeutic or material platforms.

[View in Dashboard Open Paper](#)

7. Universal rhythmic architecture uncovers two modes of neural dynamics.

28/50

"Discovering universal brain patterns enables advanced neurodiagnostics and BCIs."

Main researcher: Ayelet N Landau

Department of Psychology, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Cognitive and Brain Sciences, The Hebrew University of Jerusalem, Jerusalem, 9190401, Israel.; Department of Experimental Psychology, University College London (UCL), London, WC1H 0AP, UK.

Published in Nature communications · 2026-05

WHY NOW

This foundational insight can power next-generation brain-computer interfaces and personalized neurodiagnostic software, tapping into a rapidly growing market.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

COMMERCIALISATION METRICS

Novelty

8/10

The research shifts the fundamental paradigm of neural analysis from a frequency-only model to a two-dimensional architecture (frequency + rhythmicity), providing a potentially transformative framework for neurotechnological diagnostics. This provides a platform-enabling discovery that unifies disparate datasets across species and clinical states into a single organizing principle.

Commercial Potential

5/10

While the framework offers high potential for improving the precision of neurodiagnostic biomarkers and personalized EEG analysis, it remains a fundamental descriptive discovery rather than a specific device or therapeutic intervention. Its commercial value depends entirely on its integration into future clinical software or neurotechnology platforms.

Market Size

8/10

The findings provide a platform-enabling framework for universal neurophysiological analysis, addressing a massive market spanning clinical diagnostics, neuroprosthetics, and brain-computer interfaces across diverse patient populations.

Tech Readiness

3/10

The paper presents fundamental discovery-based neuroscience that provides a new theoretical framework for understanding neural dynamics. While it offers potential for future diagnostic biomarkers, it is currently at a basic research stage without a defined prototype, hardware integration, or clinical validation pipeline.

IP Strength

4/10

The work describes a scientific framework and a descriptive method for analyzing existing neural data rather than a novel hardware device or a proprietary biochemical entity. While the 'rhythmicity measure' could potentially be patented as a diagnostic algorithm, the primary output is foundational knowledge that is difficult to defend against competitors using alternative mathematical modeling.

[View in Dashboard Open Paper](#)

8. Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck. 28/50

"AI framework distills complex single-cell data for drug discovery."

Main researcher: Mor Nitzan · mor.nitzan@mail.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel; Racah Institute of Physics, The Hebrew University of Jerusalem, Jerusalem, Israel; Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.

Electronic address: mor.nitzan@mail.huji.ac.il.

Published in Cell systems · 2026-05

WHY NOW

Addressing the signal-to-noise challenge in genomics, bioIB provides a crucial software tool for identifying disease biomarkers and therapeutic targets efficiently.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

COMMERCIALISATION METRICS

Novelty

6/10

The paper applies an established information theory framework (Information Bottleneck) to a well-trodden domain (scRNA-seq), representing an incremental methodological improvement rather than a fundamental scientific breakthrough. While the application to gene clustering and hierarchical structure is useful, it follows existing trends in deep learning-based dimensionality reduction and feature selection.

Commercial Potential

6/10

The framework presents a strong platform-enabling computational tool for drug discovery and biomarker identification, but its commercial value is currently tied to software/algorithm licensing rather than a standalone proprietary product. Its utility depends on integration into existing bioinformatics pipelines or single-cell analysis platforms.

Market Size

8/10

The framework acts as a platform-enabling technology for the massive and growing single-cell multi-omics market, offering high value for drug discovery and precision medicine. By optimizing signal extraction in complex datasets, it addresses a critical bottleneck in the multi-billion dollar bioinformatics and clinical diagnostics sectors.

Tech Readiness

4/10

The paper presents a computational framework and algorithmic methodology (bioIB) rather than a physical product or clinical tool. While it demonstrates utility on existing datasets, it remains at the fundamental software/algorithm development stage (TRL 3-4) rather than a validated industrial platform.

IP Strength

4/10

The innovation is primarily a computational framework based on the Information Bottleneck method, which is an established theoretical principle in machine learning, making the core algorithm difficult to patent. While the application to scRNA-seq is valuable, the defensibility is low as it likely falls under 'mathematical methods' and is susceptible to rapid replication by open-source community developments.

[View in Dashboard](#) [Open Paper](#)

9. RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

"Bacterial protein activates NK cells, a new frontier in cancer immunotherapy."

Main researcher: Ofer Mandelboim

The Concern Foundation Laboratories at the Lautenberg Center for Immunology and Cancer Research, Institute for Medical Research Israel Canada (IMRIC), Hebrew University Hadassah Medical School, Jerusalem, Israel.

Published · 2026-05

WHY NOW

Uncovering a novel mechanism to boost NK cell activity, this presents an exciting opportunity for developing next-generation immunotherapies in oncology.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

COMMERCIALISATION METRICS

Novelty

7/10

The discovery of a specific bacterial protein (RadD) acting as a direct ligand for the NKp46 receptor is highly novel and expands the known ligandome of NK cells. While it provides a strong foundation for targeted immunotherapies, it is an incremental discovery of a molecular interaction rather than a platform-level technological breakthrough.

Commercial Potential

6/10

The identification of RadD as a specific ligand for NKp46 provides a clear molecular target for developing NK cell agonists or mimics to enhance antitumor cytotoxicity. However, the commercial path is complicated by the dual nature of *F. nucleatum* in cancer progression and the need for precise delivery to avoid systemic inflammation.

Market Size

6/10

The target market encompasses oncology immunotherapy, specifically head and neck cancers, which is substantial; however, the current focus is on a highly specific host-microbe interaction rather than a broad platform application.

Tech Readiness

3/10

The research is at the fundamental discovery stage, establishing a biological mechanism through in vitro assays and murine models. While it identifies a novel target for immunotherapy, there are no lead candidates, pharmacokinetic data, or clinical validation steps presented.

IP Strength

6/10

The identification of a specific bacterial ligand (RadD) for a known human receptor (NKp46) provides a clear target for composition-of-matter patents on RadD-mimetics or agonists. However, since RadD is a naturally occurring bacterial protein, the IP strength depends on the ability to engineer novel derivatives or specific delivery vehicles rather than the discovery of the axis itself.

[View in Dashboard](#) [Open Paper](#)

10. Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2 27/50

"Targeting nitric oxide offers precision therapy for autism subtypes."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

Institute for Drug Research, School of Pharmacy, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel.
Haitham.amal@mail.huji.ac.il; Rosamund Stone Zander and Hansjoerg Wyss Translational Neuroscience Center, Boston Children's Hospital, Harvard Medical School, Harvard University, Boston, MA, USA. Haitham.amal@mail.huji.ac.il.
Published in Molecular autism · 2026-04

WHY NOW

These findings provide specific molecular targets for developing small-molecule drugs for genetically defined autism, addressing a critical unmet medical need.

COMMERCIAL ANGLE

The findings support the development of small-molecule therapeutics targeting nitric oxide synthase as a precision medicine approach for genetically defined autism subtypes.

COMMERCIALISATION METRICS

Novelty

5/10

The study follows a logical progression from previously established findings (implicated NO signaling) to mechanistically characterizing those findings via proteomics. While it provides important biological validation, the innovation is incremental rather than a groundbreaking discovery of a new pathway or a novel therapeutic modality.

Commercial Potential

6/10

The study identifies a specific molecular target (nNOS inhibition) and a drug (7-NI) that shows functional efficacy in two different genetic models of ASD, providing a clear path toward drug repurposing or development. However, the commercial potential is limited by the complexity of treating neurodevelopmental disorders and the significant translational gap between mouse proteomic correction and human clinical efficacy.

Market Size

9/10

The research targets Autism Spectrum Disorder (ASD), a massive global market with high unmet medical need and a large patient population. Furthermore, the identification of a specific, conserved molecular mechanism (NO signaling) suggests a potential platform for treating multiple genetic subtypes of ASD.

Tech Readiness

3/10

The research is in the early preclinical stage, focusing on mechanistic proteomic profiling and behavioral validation in murine models. While it identifies a potential therapeutic target (nNOS inhibition), it lacks the pharmacokinetic, safety, or dose-escalation data required for clinical translation.

IP Strength

4/10

The innovation relies on 7-Nitroindazole, a well-known, off-patent small molecule, which limits the ability to claim novel composition of matter. While the proteomic targets identified could potentially support new method-of-use patents, the underlying mechanism is incremental rather than a breakthrough platform technology.

[View in Dashboard](#) [Open Paper](#)

11. Model-Based Attribution of Biologic Effectiveness in Psoriasis: Drug vs Context Contributions.

26/50

"AI model predicts biologic drug success, revolutionizing patient therapy."

Main researcher: Yuval Ramot · yuval.ramot@mail.huji.ac.il.

Faculty of Medicine, Hebrew University of Jerusalem, Jerusalem, Israel; Service for Inflammatory Diseases of the Skin, Department of Dermatology, Hadassah Medical Center, Jerusalem, Israel. Electronic address: yuval.ramot@mail.huji.ac.il.

Published in Journal of the American Academy of Dermatology · 2026 Jul 15

WHY NOW

This precision medicine software can reduce trial-and-error in prescribing, leading to better patient outcomes and significant cost savings in immunology.

COMMERCIAL ANGLE

Development of a precision medicine software tool for predicting patient response to biologic therapies, reducing trial-and-error prescribing in immunology.

COMMERCIALISATION METRICS

Novelty

6/10

The approach of using model-based attribution to decouple drug effects from patient context is a sophisticated application of causal inference/ML, but it builds on existing attribution frameworks rather than inventing a new scientific paradigm.

Commercial Potential

6/10

The work suggests a high-value application in precision medicine and companion diagnostics for biologics, but without the abstract, it is unclear if this is a general theoretical framework or a scalable, proprietary tool.

Market Size

8/10

Psoriasis is a high-prevalence chronic condition with a massive, high-value biologics market; tools that optimize drug effectiveness and patient stratification represent a significant commercial opportunity.

Tech Readiness

3/10

The title indicates a model-based theoretical or analytical framework for attributing drug effectiveness, which is characteristic of early-stage computational research (TRL 3) rather than a validated clinical tool or product.

IP Strength

3/10

The title suggests a methodological or diagnostic framework for attribution analysis, which is typically difficult to patent and often categorized as a mathematical method or software tool with low inherent defensibility.

[View in Dashboard](#) [Open Paper](#)

12. Strategic Template Filtering Accelerates Fragment-Based Peptide Docking. 26/50

"Faster peptide docking software accelerates AI-driven drug discovery."

Main researcher: Ora Schueler-Furman

Microbiology and Molecular Genetics & Quantitative Molecular Medicine, The Hebrew University of Jerusalem, Jerusalem 91120, Israel.

Published in Journal of chemical information and modeling · 2026-06

WHY NOW

By dramatically improving efficiency without sacrificing accuracy, this algorithm enhances the throughput of AI platforms in designing peptide therapeutics.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

COMMERCIALISATION METRICS

Novelty

4/10

The work describes an incremental optimization of an existing protocol (PatchMAN) by adding filtering and localization to reduce compute time. It is an efficiency improvement rather than a groundbreaking shift in methodology or a novel scientific discovery.

Commercial Potential

6/10

The tool provides a clear efficiency gain for an existing pipeline, making peptide-protein docking more accessible for drug discovery. However, it is an incremental optimization of a protocol rather than a novel, standalone platform or a direct therapeutic lead.

Market Size

8/10

Peptide-protein interaction modeling is a foundational platform technology for the multi-billion dollar biologics and drug discovery markets. Reducing the computational cost of high-accuracy docking directly enables larger-scale virtual screening and accelerates the development of peptide-based therapeutics.

Tech Readiness

4/10

The work is a computational optimization of an existing software protocol, moving it from a high-cost academic tool to a more efficient one, but it remains in the 'tool development' stage without evidence of integrated drug discovery pipeline application.

IP Strength

4/10

The innovation is an incremental optimization of an existing open-source protocol (PatchMAN) focused on computational efficiency rather than a novel, proprietary algorithm. As a software refinement for fragment filtering, it is likely viewed as an obvious improvement in the field, making it difficult to secure strong, broad patents.

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